

Q10

Identities of sets:

$$(1) \left. \begin{aligned} A \cup \emptyset &= A \\ A \cap U &= A \end{aligned} \right\} \text{identity law}$$

$$(2) \left. \begin{aligned} A \cup U &= U \\ A \cap \emptyset &= \emptyset \end{aligned} \right\} \text{Domination law.}$$

$$(3) \left. \begin{aligned} A \cup A &= A \\ A \cap A &= A \end{aligned} \right\} \text{Idempotential law.}$$

$$(4) (\bar{\bar{A}}) = A \rightarrow \text{Complementation law.}$$

$$(5) A \cup B = B \cup A \rightarrow \text{Commutative law.}$$

$$(6) A \cap B = B \cap A \rightarrow \text{Commutative law.}$$

$$(6) \left. \begin{aligned} A \cup (B \cap C) &= (A \cup B) \cap C \\ A \cap (B \cup C) &= (A \cap B) \cup C \end{aligned} \right\} \text{Associative laws}$$

$$(7) \left. \begin{aligned} A \cap (B \cup C) &= (A \cap B) \cup (A \cap C) \\ A \cup (B \cap C) &= (A \cup B) \cap (A \cup C) \end{aligned} \right\} \text{Distribution law}$$

$$(8) \begin{cases} \overline{A \cup B} = \bar{A} \cap \bar{B} \\ \overline{A \cap B} = \bar{A} \cup \bar{B} \end{cases} \text{De-morgan's laws}$$

$$(9) \begin{cases} A \cup (A \cap B) = A \\ A \cap (A \cup B) = A \end{cases} \text{Absorption laws}$$

Morgan's law: $\overline{A \cap B} = \bar{A} \cup \bar{B}$ [by def. of complement]

$$\therefore \overline{A \cap B} = \{x \mid x \notin A \cap B\}$$

$$= \{x \mid \neg (x \in (A \cap B))\}$$

$$= \{x \mid \neg (x \in A \wedge x \in B)\}$$

$$= \{x \mid \neg (x \in A) \vee \neg (x \in B)\}$$
 [by def. of De Morgan's law for logical equivalence]

$$= \{x \mid x \notin A \vee x \notin B\}$$

$$= \{x \mid x \in \bar{A} \vee x \in \bar{B}\}$$

$$= \{x \mid x \in (\bar{A} \cup \bar{B})\}$$
 [by the def of union]

$$= \bar{A} \cup \bar{B}$$

• All side notes must

(*) Show that, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ with membership table.

A	B	C	$B \cup C$	$A \cap B$	$A \cap C$	$A \cap (B \cup C)$	$(A \cap B) \cup (A \cap C)$
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0
1	1	0	1	1	0	1	1
1	1	1	1	1	1	1	1

Hence, the two sides are equal.
 $\therefore A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$